A Decision Tree Model for Predicting Student Performance in the Common Entrance Examination (CEE)

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Abstract

This study applies a Decision Tree Classifier to predict student performance in the Common Entrance Examination (CEE) using selected demographic, academic, and socioeconomic factors. By analysing key attributes such as gender, time spent studying, medium of instruction, academic percentages, and parental occupation, the model aims to assist educational institutions in identifying students who may require academic support or intervention.

1 Problem Definition

The primary goal of this research is to predict whether a candidate will perform well in the Common Entrance Examination (CEE) using features from their academic history and demographic information. The dataset includes:

- Gender of the candidate
- Time spent studying
- Medium of instruction in Class XII
- Percentage in Class X
- Percentage in Class XII
- Father's occupation
- Mother's occupation

2 Relevance in the Philippine Setting

Although the dataset originates from an external educational system, the methodology is highly applicable in the Philippines:

- Filipino students take entrance exams for colleges, STEM programs, scholarships, and government institutions.
- Socioeconomic factors (parents' occupation, study time, school instruction medium) strongly influence academic performance in the Philippines.
- Predictive models can help CHED, DepEd, and LGUs design academic support programs.

3 Problem Type

This is a **Classification Problem**, where the task is to predict a performance class (e.g., Excellent, Good, Average, Poor).

4 Proposed Algorithm: Decision Tree Classifier

Justification

A Decision Tree is selected because:

- It handles categorical and numerical features without heavy preprocessing.
- It produces interpretable rules, making it easy for educators and policymakers to understand.
- It shows which features most strongly predict exam performance.
- It performs well on moderate-size educational datasets.

Secondary models (optional comparison):

- Logistic Regression
- Random Forest (for potential higher accuracy)

5 Methodology

5.1 Data Acquisition and Provenance

The dataset used is `CEE_DATA.csv`. To prove Philippine provenance (if applicable), attach:

- Direct source URL from a Philippine institution (DepEd, CHED, university registrar).
- Metadata or documentation describing Philippine collection.
- Official acknowledgement from the institution providing the dataset.

5.2 Data Preprocessing

1. Load CSV and inspect missing values.
2. Encode categorical features using LabelEncoder.
3. Ensure percentage scores are numeric.

5.3 Feature Selection

Selected features for this study:

- Gender
- Time spent studying
- Medium of instruction
- Class X Percentage
- Class XII Percentage

- Father's occupation
- Mother's occupation

Excluded features:

- Caste
- Coaching attendance
- Class X board
- Class XII board

5.4 Model Training

1. Train-test split: 70/30.
2. Train the Decision Tree Classifier:

```
from sklearn.tree import DecisionTreeClassifier
model = DecisionTreeClassifier(random_state=42)
model.fit(X_train, y_train)
```

5.5 Model Evaluation

Metrics:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

5.6 Interpretation

- Identify which features most influence prediction.
- Extract decision rules (tree structure).
- Evaluate fairness (e.g., bias in gender or socioeconomic variables).

6 Dataset Features and Target

Features:

- Gender, Time, Medium, Class X Percentage, Class XII Percentage, Father Occupation, Mother Occupation

Target Variable:

- `Performance` (classes: Excellent, Good, Average, Poor)

7 Dataset Size

After running the inspection code (Appendix A):

- Number of samples: **¡REPLACE¿**
- Number of usable features (after encoding): **7**

8 Intended Output

The project will deliver:

- A Decision Tree model capable of predicting student performance.
- A simple interface (notebook or web app) that accepts attributes and returns predictions.
- Feature importance ranking for educational insights.

9 Philippine Stakeholder Applications

- **Universities**: Identify at-risk applicants early.
- **Local Governments**: Use model insights for scholarship screening.
- **DepEd/CHED**: Evaluate the effect of socioeconomic variables on learning outcomes.

References

- Quinlan, J.R. (1986). Decision Trees and ID3.
- DepEd/CHED education statistics portals.